

4. Project Planning Phase

ComicCraft - AI/ML & Generative AI Project

1. Development Plan

- Finalize project problem statement and objectives.
- Analyze functional and non-functional requirements.
- Design the application architecture and user flow.
- Develop the FastAPI backend.
- Integrate Gemini AI for Generative AI content generation.
- Develop and connect the frontend interface.
- Test the application and fix integration issues.
- Prepare project documentation and demonstration.

2. Project Milestones

Milestone	Planned Activity
M1 - Ideation	Brainstorming, problem identification and idea selection.
M2 - Requirements	Define functional, non-functional, hardware and software requirements.
M3 - Design	Create system architecture, modules and user flow.
M4 - Development	Implement FastAPI, frontend and Gemini AI integration.
M5 - Testing	Test startup, API communication, generation and output display.
M6 - Documentation	Prepare phase-wise project documents and technical documentation.
M7 - Demonstration	Demonstrate prompt input, AI generation and four-panel result.

3. Resource Planning

- Development machine or laptop with Python support.
- Internet connection for Generative AI API communication.
- Python development environment and virtual environment.
- FastAPI framework and required Python packages.
- Gemini AI API credentials stored through environment variables.
- Modern browser for frontend testing.

4. Risk Management Plan

Risk	Mitigation
Missing or invalid API configuration	Check environment variables and validate configuration before generation.

API/network failure	Handle exceptions and display a clear error message to the user.
Unexpected AI response	Use structured prompts and validate the returned content before display.
Integration mismatch	Keep frontend request fields and backend response fields consistent and test the API independently.
Future image-generation latency/cost	Treat visual panel generation as an optional extension and optimize requests.

5. Timeline

The project follows a phase-based development approach: ideation -> requirements -> design -> planning -> development -> testing -> documentation -> demonstration.

6. Deliverables

- Phase-wise PDF documentation
- Working ComicCraft web application
- Generative AI integration using Gemini
- Four-panel text-based comic output
- Testing records
- Project demonstration material

7. Future Plan

After the core text-generation workflow, the next development stage can add image generation, character consistency, speech bubbles, comic styling and PDF/image export.